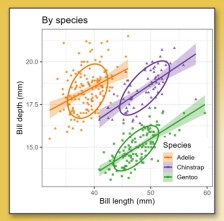
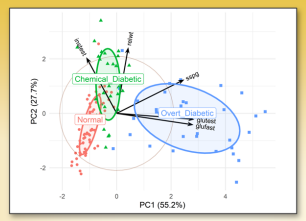
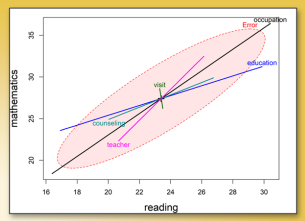






Visualizing Multivariate Data and Models in R

A Romance in Many Dimensions

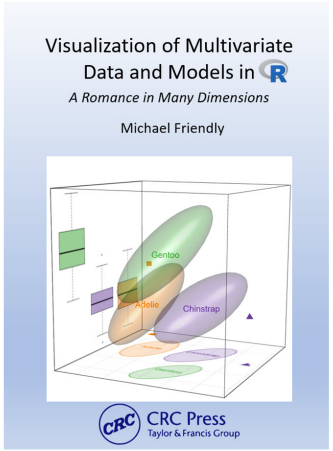
Methodology & Statistics Group, Utrecht, Nov. 2026
Michael Friendly, York University, friendly.github.io | datavis.ca

1

My book

Graphical methods for multivariate data and for multivariate linear models

- 15 chapters in four parts, from exploratory plots to MANOVA and discriminant analysis
- To appear CRC Press / Chapman & Hall, 2027
- Online version: friendly.github.io/Vis-MLM-book



2

2

R packages

Theory → Practice: Making tools accessible

heplots
Visualizing hypothesis tests in multivariate linear models: HE plots, covariance ellipses, robust MLMs

candisc
Canonical discriminant and canonical correlation analysis: low-dimensional views of multivariate effects

mvinfluence
Influence measures and diagnostic plots for multivariate linear models

genridge
Generalized ridge trace plots: bias and precision in ridge regression

VisCollin
Understanding problems of collinearity thru visualization and reduced-rank views

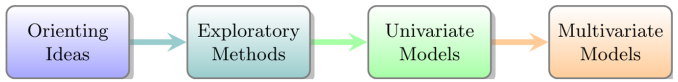
3

3

Ideas I'll talk about today (thematic)

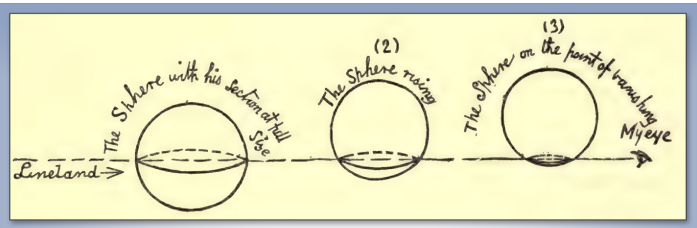
- Why a romance in many dimensions?
- Seeing multivariate data: ellipses everywhere
- Multivariate juicers: low-D views, and Waldo space
- Elliptical insights: models with one response
- The HE plot framework for multivariate models
- Checking multivariate models

Road map of the book (linear)



4

4



Why a romance in many dimensions?

How can we understand important scientific phenomena?

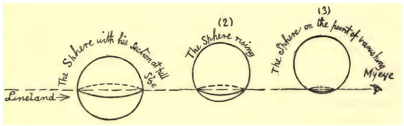
Our data live in Spaceland, sometimes high-D, but we plot in Planeland

5

5

Planeland: we only ever see 2D projections

- Abbott's Flatland (1884): a sphere passing through a 2D world is seen as a circle that grows and shrinks
- Our 2D plots are Flatland views of data that live in many dimensions
- Multivariate visualization is the art of **choosing good views**
- #dataviz is the art of making plots **informative** (& pretty!)
- **Truth & Beauty**: That's the Romance!



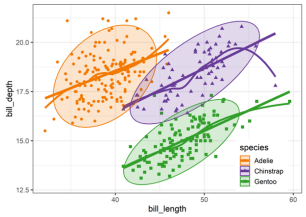
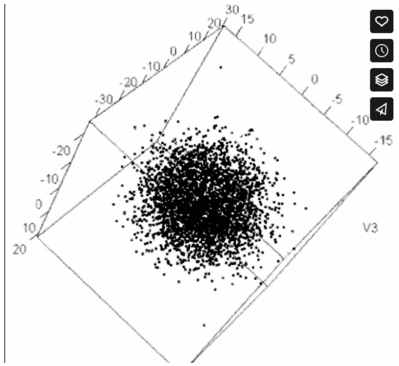


Figure 1.1

6

6

EUREKA! Structure can hide in many dimensions



HistData::Pollen, 5D, from the 1986 JSM Data Exposition: "Find something interesting here"

Zooming in on the **right view** reveals what no summary statistic could ever do

History of science gives many Eureka Moments revealed by graphs

Figure 1.5

7

7

Multivariate discovery: kinds of Type 2 diabetes

- Reaven & Miller (1979) viewed glucose and insulin measures in 3D with PRIM-9
- Earlier 1968 study: murky in 2D
- The 3D view showed distinct groups that 2D plots had obscured
- The picture changed how the disease was classified:
  - Chemical (Prediabetes / Early Type 2)
  - Overt (Advanced Type 2)

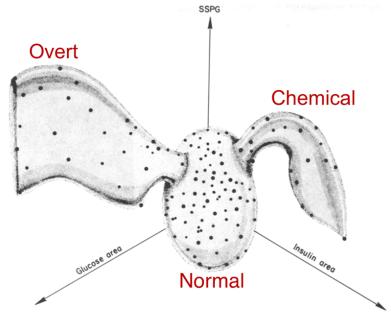


Figure 1.7

8

8

MLM problem: easy to fit, hard to see

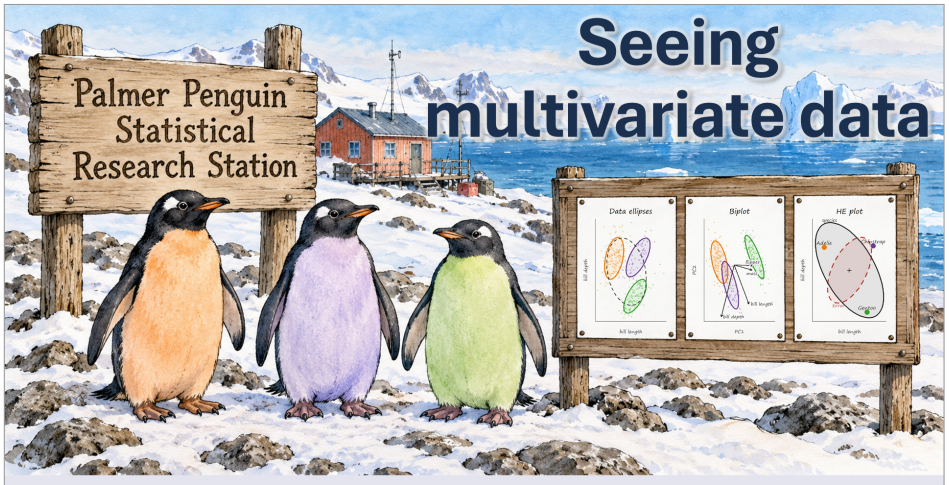
- With one response, we have a rich toolkit of plots for linear models
- With several responses, `lm(cbind(y1,y2,y3) ~ ...)` is just as easy to fit
- Math is beautiful & easy too: $y \rightsquigarrow \mathbf{Y}$
- But the results arrive as tables of multivariate tests 😟
- Goal: plots that show what those tests mean, visual inference? 😇

Response variables: $\mathbf{Y} = (y_1, \ldots y_q)$

Predictor variables: $\mathbf{X} = (x_1, \ldots x_p)$		Quantitative		Discrete	
		q=1	q>1	q=1	q>1
Quantitative	p=1	Simple regression	Multivariate regression	Simple logistic regression	
	p>1	Multiple regression	Multivariate regression Canonical corr. Partial corr.	Mult. logistic regression Discriminant analysis	Multivariate logistic regression
Discrete	p=1	t-test 1-way ANOVA	Hotelling T^2 1-way MANOVA	Simple χ^2	Loglinear models
	p>1	Factorial ANOVA	Factorial MANOVA	Logit models Loglinear models	Loglinear models

Figure 6.1

9



Data ellipses, penguins, and looking at many variables at once

10

9 10

Galton's visual discovery:

Bivariate normal correlation surface (1886)

Table 9.1 One of Galton's correlation tables

Height of the mid-parent in inches	Height of the adult child													
	<61.7	62.2	63.2	64.2	65.2	66.2	67.2	68.2	69.2	70.2	71.2	72.2	73.2	>73.7
>73.0	—	—	—	—	—	—	—	—	—	—	—	1	3	—
72.5	—	—	—	—	—	—	—	1	2	1	2	7	2	4
71.5	—	—	—	—	1	3	4	3	5	10	4	9	2	2
70.5	1	—	1	—	1	1	3	12	18	14	7	4	3	3
69.5	—	—	1	16	4	17	27	20	33	25	20	11	4	5
68.5	1	—	7	11	16	25	31	34	48	21	18	4	3	—
67.5	—	3	5	14	15	36	38	28	38	19	11	4	—	—
66.5	—	3	3	5	2	17	17	14	13	4	—	—	—	—
65.5	1	—	9	5	7	11	11	7	7	5	2	1	—	—
64.5	1	1	4	4	1	5	5	—	2	—	—	—	—	—
<64.0	1	—	2	4	1	2	2	1	1	—	—	—	—	—
Totals	5	7	32	59	48	117	138	120	167	99	64	41	17	14
Medians	—	—	66.3	67.8	67.9	67.7	67.9	68.3	68.5	69.0	69.0	70.0	—	—

Source: Galton (1886), p. 68.

11

Visual smoothing → Insight

Table 9.1 One of Galton's correlation tables

Height of the mid-parent in inches	Height of the adult child													
	<61.7	62.2	63.2	64.2	65.2	66.2	67.2	68.2	69.2	70.2	71.2	72.2	73.2	>73.7
>73.0	—	—	—	—	—	—	—	—	—	—	—	1	3	—
72.5	—	—	—	—	—	—	—	1	2	1	2	7	2	4
71.5	—	—	—	—	1	3	4	3	5	10	4	9	2	2
70.5	1	—	1	—	1	1	3	12	18	14	7	4	3	3
69.5	—	—	1	16	4	17	27	20	33	25	20	11	4	5
68.5	1	—	7	11	16	25	31	34	48	21	18	4	3	—
67.5	—	3	5	14	15	36	38	28	38	19	11	4	—	—
66.5	—	3	3	5	2	17	17	14	13	4	—	—	—	—
65.5	1	—	9	5	7	11	11	7	7	5	2	1	—	—
64.5	1	1	4	4	1	5	5	—	2	—	—	—	—	—
<64.0	1	—	2	4	1	2	2	1	1	—	—	—	—	—
Totals	5	7	32	59	48	117	138	120	167	99	64	41	17	14
Medians	—	—	66.3	67.8	67.9	67.7	67.9	68.3	68.5	69.0	69.0	70.0	—	—

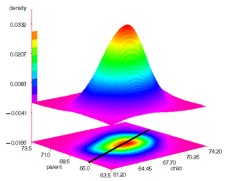
Source: Galton (1886), p. 68.

12

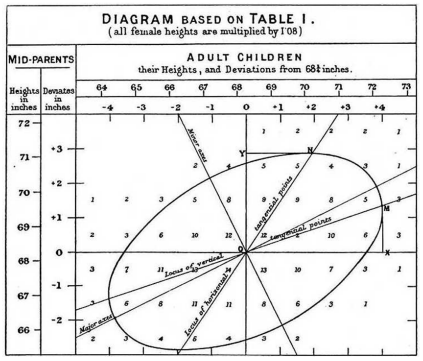
11 12

Visual insight → Theory (the OMG! moment)

- 3D Level curves are **ellipses**
- Regression lines are loci of conjugate **tangents**
- PCA dims are **major / minor axes**



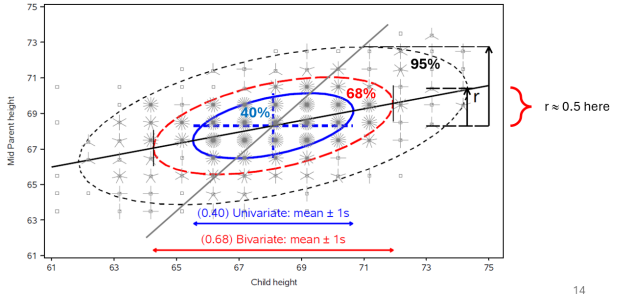
... that Galton should have evolved all this ... is to my mind one of the most note-worthy scientific discoveries arising from analysis of pure observation (Pearson 1920, p37)



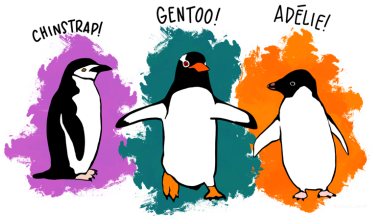
Galton (1886, Pl X): Smoothed contours of heights of parents and children

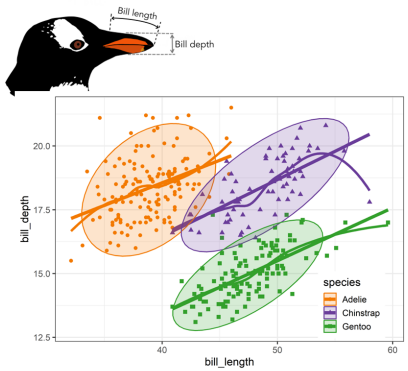
The data ellipse summarizes means, spread & correlation – a **sufficient visual summary** for *N*(μ, Σ)

- Ellipsoid boundary: Mahalanobis $D_M^2(y_i) \sim \chi^2_p$ $\quad D^2_M(\mathbf{y}) = (\mathbf{y}-\bar{\mathbf{y}})^\mathsf{T} S^{-1} (\mathbf{y}-\bar{\mathbf{y}})$
  - *p*=2: shadows generalize univariate confidence intervals
  - eccentricity: precision → visual estimate of correlation



Meet the penguins





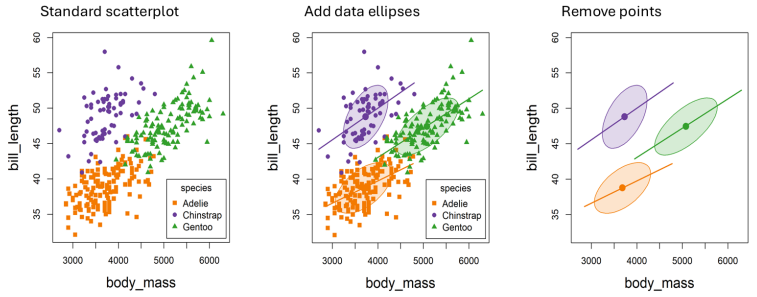
Three species, four size measures: bill length, bill depth, flipper length, body mass. Artwork: Allison Horst.

Figure 4.17, 4.19

Data ellipses: Sufficient visual summary

Plot annotations are **visual summaries** (ellipse, regression line, conf. band)

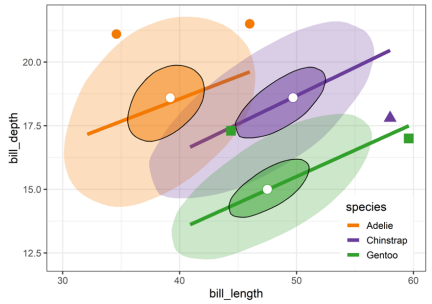
- Add **data ellipses** to show w/in group means & scatter (sufficient for N(**μ, Σ**))
- Remove the points to show the overall **patterns**: means, shape, corr., precision
- What remains is **all that linear models care about** (ellipses can be made robust!)



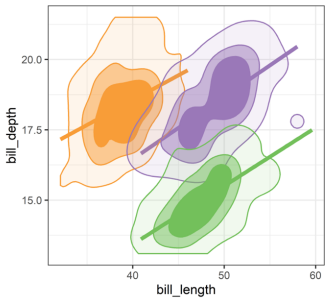
What if I'm not Normal? 😳

Don't worry: there are non-parametric alternatives

`gggda::geom_bagplot()`



`ggdensity::geom_hdr()`

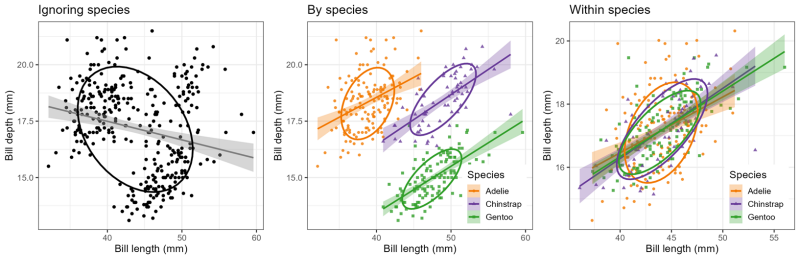


Bags and high-density regions don't assume normality, yet for the penguins they come out nearly elliptical

Figure 4.21, 4.22

17

Simpson's paradox: marginal & conditional relations



Ignoring species, longer bills are shallower

By species, relation is positive for each

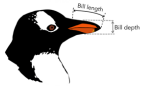
Within species shows diffs in slopes & covariances

Figure 4.23

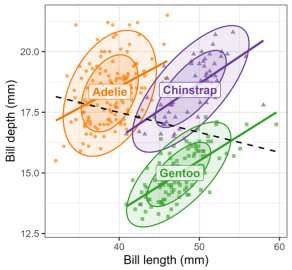
18

Thinking about Penguin variables

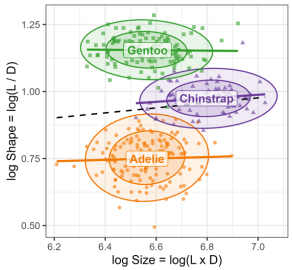
- Bill length & depth are two measurements of one object: a penguin's **beak**.
- → Re-express them as **Size** = L x D and **Shape** = L / D
- On log scales, a **45°** rotation: equivalent data → same multivariate tests
- Simpson's paradox **disappears**: species differ in **shape**; within species, bills vary in **size**



Length & depth



Size & shape



Blog: friendly.github.io/blog/posts/2026-10-peng-altviews

19

How to visualize $p > 2$ variables in Planeland?

Scatterplot matrices: all pairs at once

- Every pairwise view in one display
- Data ellipses & regression lines by species
- Works well up to a moderate number of variables
- **Visual thinning**: data → ellipses + regression lines

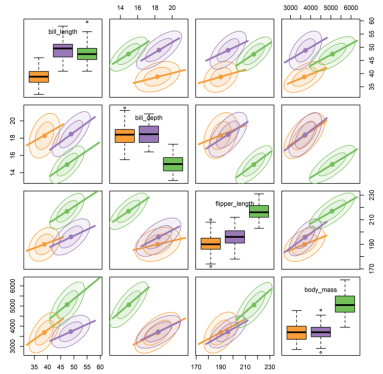
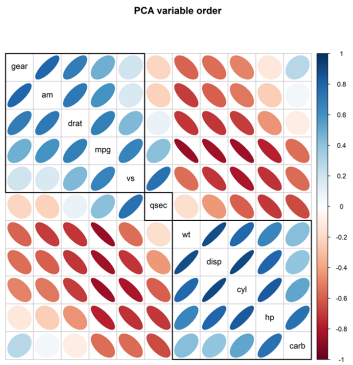


Figure 4.31

20

Visual thinning: show less to see more
Effect order: order variables to see what you want to show

- `corrplot()`: With many variables, replace the points by a rendering of each correlation
- Order the variables so that similar ones are adjacent
- Here: ordered by the angles of the variable vectors from a PCA



Friendly & Kwan (2003). Effect ordering for data displays. *Computational Statistics & Data Analysis*, *43*, 509–539. https://doi.org/10.1016/S0167-9473(02)00290-6

21

21

Parallel coordinates: one line per penguin

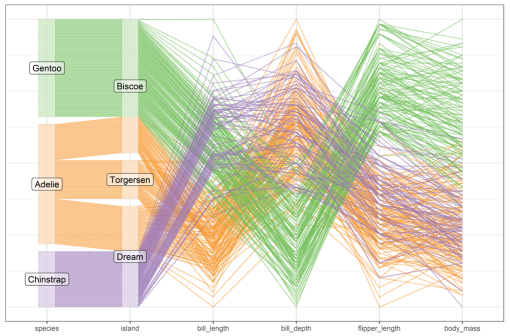


Figure 4.40

Each observation is a line across the variable axes (`Ggally`)
- Groups appear as bundles
- Easy to extend to more variables

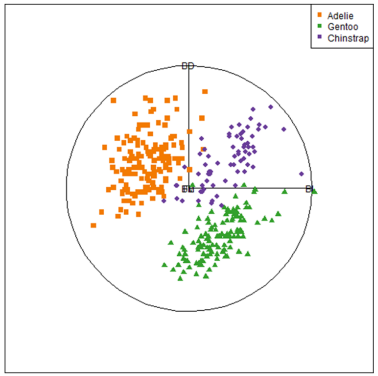
Variable ordering is important here → We see that penguins like particular islands

Using this interactively could make a tool for understanding (`plotly`)

22

22

Animated tours: flying through projections (**`tourr`** pkg)



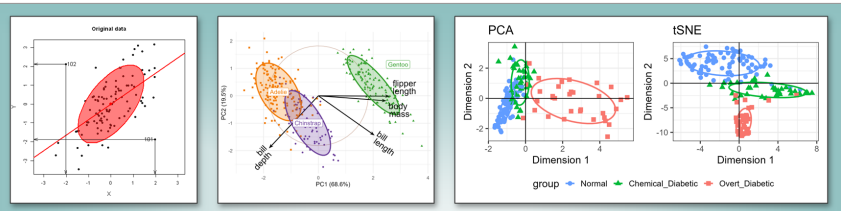
Animation adds one more dimension to what we can see from 2D projections

A **grand tour** shows a smooth sequence of 2D projections
- Variable vectors show where you are in n-D space (as in biplot)

A **guided tour** steers toward interesting ones.
- Optimize an index: separation of groups, anomalies (outliers)

23

23



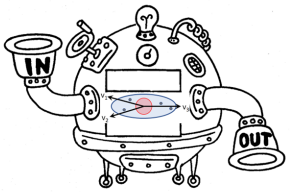
Flatland and Spaceland: dimension reduction

PCA, biplots, and nonlinear methods

24

24

Multivariate Juicers



Squeeze the most flavor from your data in low-D space

- Biplot: The data juicer
- CDA plots: The MANOVA juicer

25

PCA: rotate to the views that show the most variance

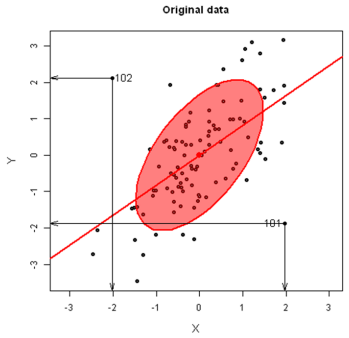


Figure 5.35

PCA is a rotation of the axes:

- PC1 lies along the direction of greatest variance,
- PC2 is orthogonal to it.

Outliers typically appear on the **smallest** dimensions

26

PCA fitted by springs: Physical analog

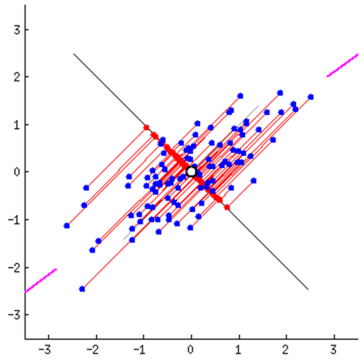


Figure 5.5

Least squares ↔ observations connected by springs

For PCA, connect a rigid line to the points by springs, always ⊥ line

Springs pull the line ∝ squared distance

It settles on the first principal component: largest variance

→ Statistics, dataviz, geometry & physics are tightly connected

27

Biplots: observations and variables in one view

- **Points**: observations (US states) in the PC1-PC2 plane
- **Vectors**: variables (crime rates)
  - **Angles** between vectors approximate correlations
  - **Projecting** a point on a vector approximates its value on that variable
- Interpretation:
  - PC1: Overall crime rate
  - PC2: Personal vs. property crime
  - Region: summarizes diffs

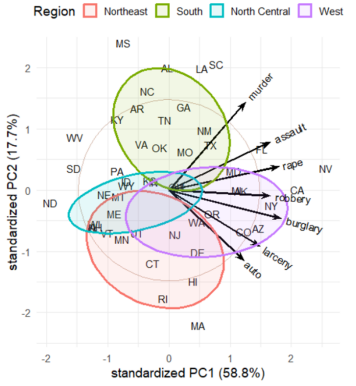


Figure 5.14

28

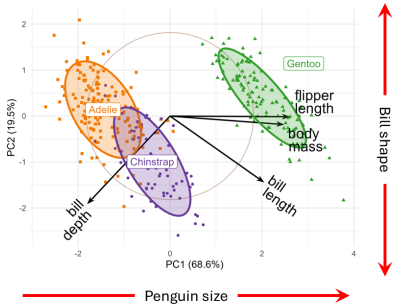
Biplot: The data juicer

```
peng.pca <- prcomp (~ bill_length + bill_depth + flipper_length + body_mass)
ggbiplot(peng.pca, ...)
```

PC1, PC2 ~ 88.1% of variance ✓

- PC1: largely flipper length & body mass: "penguin size"
- PC2: relates to "bill shape"

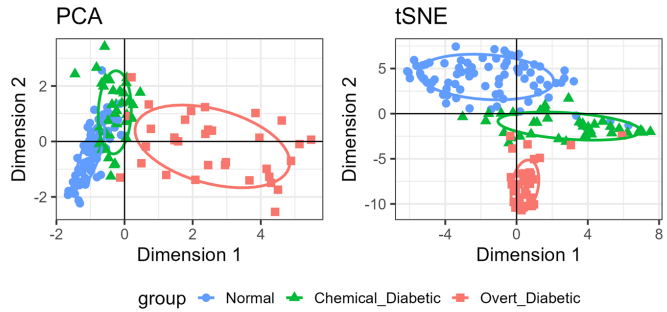
Easy to characterize the species in terms of these variables



See: https://rpubs.com/friendly/penguin-biplots

29

When a linear view is not enough: PCA vs t-SNE



t-SNE separates the Diabetes groups more cleanly, but distances & axes no longer have a direct meaning.

Figure 5.25

30

From PCA to t-SNE, animated

We can better understand algorithms visually by animating a **smooth interpolation** from one to another

Here, the PCA configuration morphs into the t-SNE one largely by a rotation.

But t-SNE achieves wider separation among groups.

The data ellipses here carry the visual message.

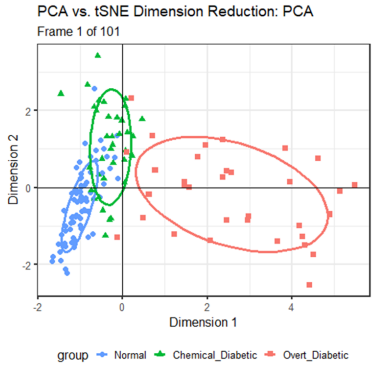
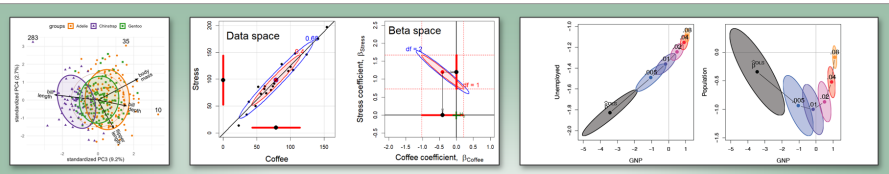


Figure 5.26

31



Elliptical Insights

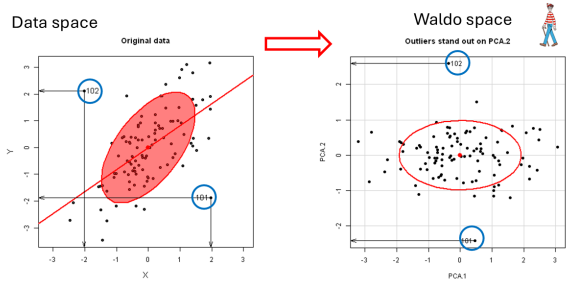
Visual understanding of problems in statistics that only confound us in tables & formulas

- Outlier detection
- Measurement error
- Understanding collinearity
- Ridge regression

32

Outlier detection: Waar is Wally?

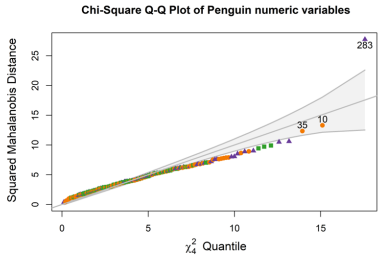
- Multivariate outliers might be invisible in univariate views, but become readily apparent on the *smallest* principal component(s)
  - 100 observations on two correlated normal variables with two bivariate outliers near (2,2), (-2, -2)



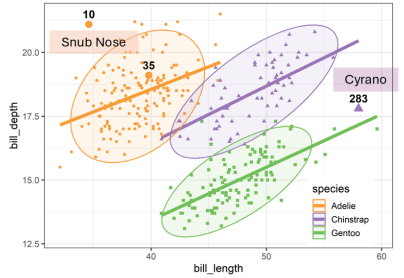
Animation: http://www.datavis.ca/gallery/animation/outlier-demo/

33

Multivariate outliers: which penguins are unusual?



Mahalanobis D^2 distances against χ^2 quantiles shows which ones are unusual



We can try to understand why they are unusual in data space of some variables

Blog: *Palmer Penguins Cry Out: "Call Me by My Name"* https://friendly.github.io/blog/posts/2026-10-peng-names/

Figure 4.26, 4.27

34

Outliers hide in the smallest dimensions

- The first two dimensions show the typical penguins
- The last dimensions show what remains
- Biplot of dimensions 3-4 of the penguin data
  - Variable vectors show relations to dimensions
  - Ellipses show diffs among groups
- Outliers are often extremes in Waldo space
  - → Biplot: why they're weird

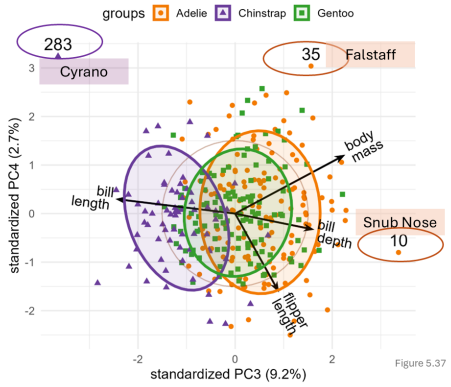


Figure 5.37

35

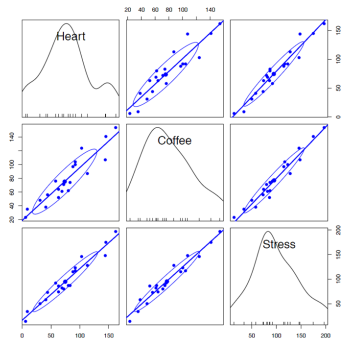
Elliptical insights: Measurement error

- In classical linear models, predictors (X) are usually assumed to be fixed (non-random), or measured w/o error
  - Rarely true in social science, medicine, …
  - (Structural equation models often used to account for this)
  - What effects do errors in **predictors** have on typical regression models?
- Main ideas:
  - Ellipses in data space show effects on bias and precision (variance^{-1})
  - The same effects can be seen in parameter (**β**) space

Friendly, Monette & Fox (2013). Elliptical Insights: Understanding Statistical Methods Through Elliptical Geometry. *Statistical Science*, *28*, 1–39. https://www.datavis.ca/papers/ellipses-STS402.pdf

36

Coffee, stress and heart disease



Imagine a small study investigating the relation between a measure of heart disease (y), and coffee consumption (x_1) and stress (x_2)

Results: lm(Heart ~ Coffee + Stress)

Coefficients and tests for the joint model predicting heart disease from coffee and stress

	Estimate ($\hat{\beta}$)	Std. error	t value	Pr(>|t|)	
Intercept	−7.7943	5.7927	−1.35	0.1961	
Coffee	−0.4091	0.2918	−1.40	0.1789	×
Stress	1.1993	0.2244	5.34	0.0001	✓

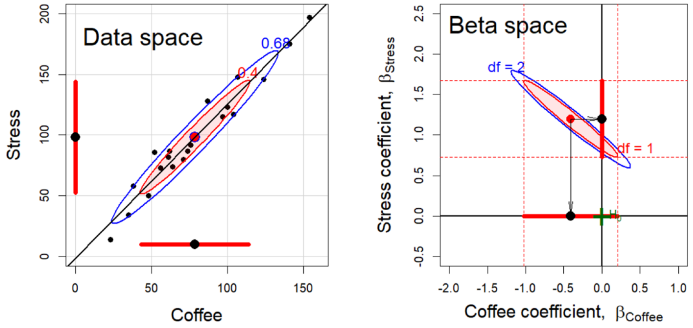
Wow! That means I can drink all the coffee I want as long as I avoid stress.

But what happens if there is measurement error in y or x_1 , x_2 ?

37

37

Data space and beta space are dual



The **data ellipse** for the predictors (**X'X**) is the inverse of the **confidence ellipse** for the coefficients, $(\mathbf{X'X})^{-1}$

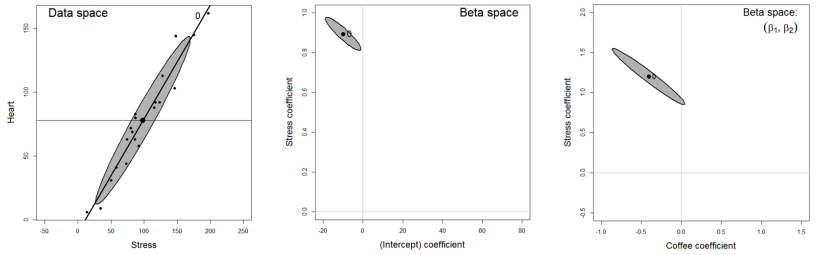
Figure 8.3

38

38

Adding measurement error

- Measurement error in **Heart** (y) decreases precision (↑ std. errors)
- Measurement error in **Stress** (x_2) biases its effect $\beta_{Stress} \rightarrow 0$ (middle)
- With Coffee in the model, it also drives β_{Coffee} toward its marginal value (right)

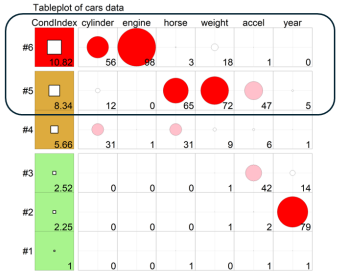


39

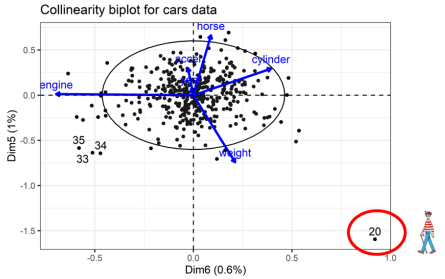
39

Understanding collinearity

Visualizing VIFs, condition indices, variance %



Collinearity problems live in Waldo space



→ Engine size too highly correlated with #cylinders

Figure 9.6, 9.7

40

40

Ridge regression: shrink to gain precision

- Penalize large coefficients

$$\text{RSS}(k) = (\mathbf{y} - \mathbf{X}\beta)^\mathsf{T}(\mathbf{y} - \mathbf{X}\beta) + k\beta^\mathsf{T}\beta$$

- The estimate moves from the OLS solution toward the origin

$$\widehat{\beta}_k^{\text{RR}} = (\mathbf{X}^\mathsf{T}\mathbf{X} + k\mathbf{I})^{-1}\mathbf{X}^\mathsf{T}\mathbf{y}$$

- A little bias buys a lot of precision
- But how can we see this?

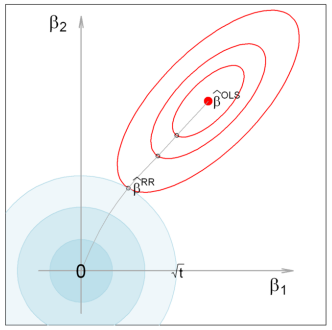


Figure 9.9

41

The usual ridge trace shows only the estimates

- Longley data: six highly collinear economic predictors
- Coefficients plotted against shrinkage constant (k)
- Shows the bias, but nothing about precision
- The **wrong plot** for a **multivariate** problem

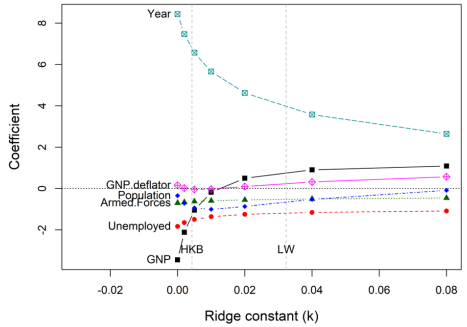


Figure 9.10

42

Generalized ridge trace plots

Rather than plotting just the univariate trajectories of β_k vs. k, plot the covariance ellipses of $\Sigma_k \equiv \text{Var}(\beta_k)$ over the range of k

- Centers of the ellipses are β_k – same info as in the univariate plot
- Can see how change in one coefficient is related to changes in others
- Relative size & shape of ellipsoids shows directly the effect on precision

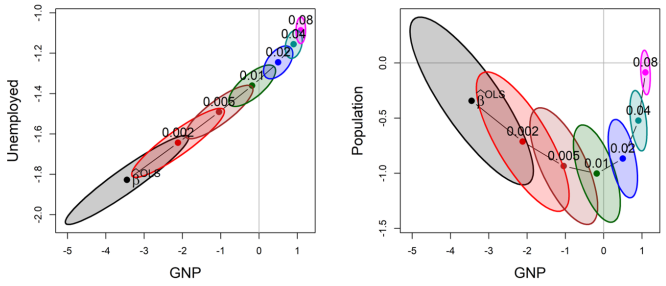


Figure 9.12

43

A low-rank view of ridge regression

- Transform to principal component space of the predictors
- → Shrinkage happens mostly in the smallest dimensions

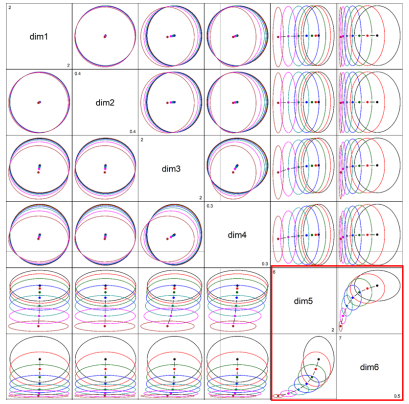


Figure 9.16

44

A low-rank view of ridge regression

- Transform to the principal component space of the predictors
- Shrinkage happens mostly in the smallest dimensions
- A biplot shows how the variables relate to them

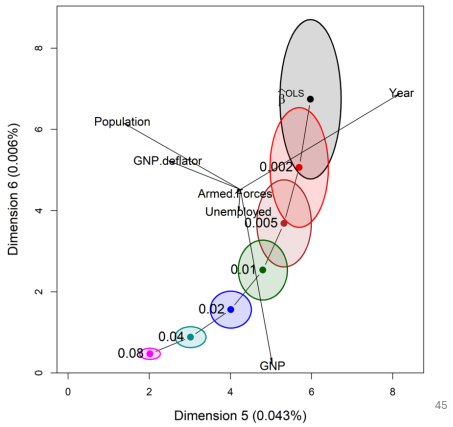
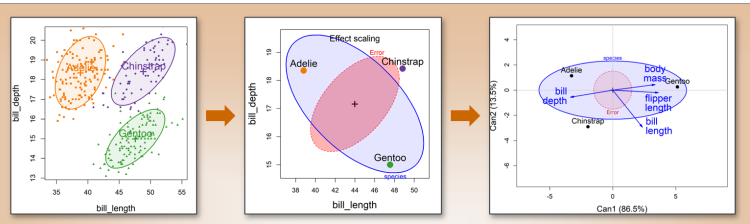


Figure 9.18

45



The HE plot framework

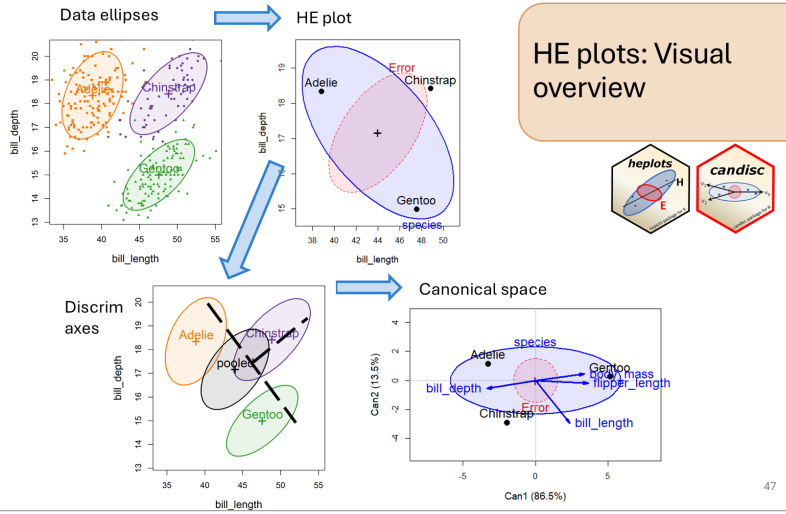
Seeing multivariate tests as ellipses for **H** and **E**

Significance scaling, seeing contrasts

Canonical space low-D views

Discriminant analysis views

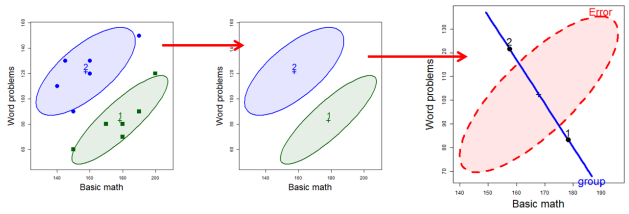
46



HE plots: Visual overview

47

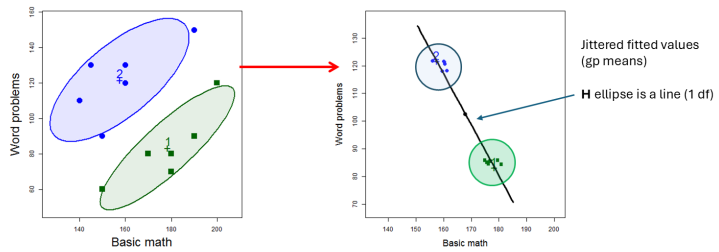
Data → Data ellipses → HE plot



- Differences **between group** means are shown by the **H** ellipsoid– data ellipsoid of the fitted values (w/ 1 df, degenerates to a line)
  - Direction shows relation of group means to response variables
  - Size shows "**how big is H relative to E**" – that's all that Wilks' Λ and other tests care about
- Variation **within groups** is reflected in the **E** ellipsoid-- data ellipsoid of the residuals
  - Direction: residual (partial) correlation between BM & WP
  - Size/shape: residual variance

48

The H ellipse



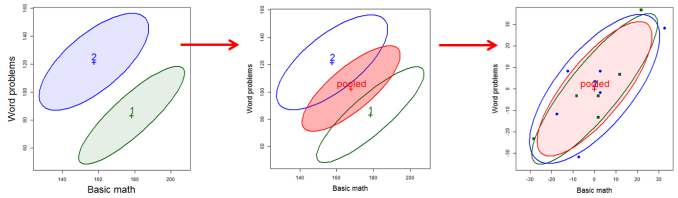
- The **H** ellipse is the **data ellipse of the fitted values** (group means, here)
  - The **H** matrix is the sum of squares and crossproducts of the fitted values, corrected for the grand mean

$$\mathbf{H} = \left(\hat{\mathbf{Y}}'\hat{\mathbf{Y}} - n\bar{\mathbf{y}}\bar{\mathbf{y}}'\right)$$

49

49

The E ellipse



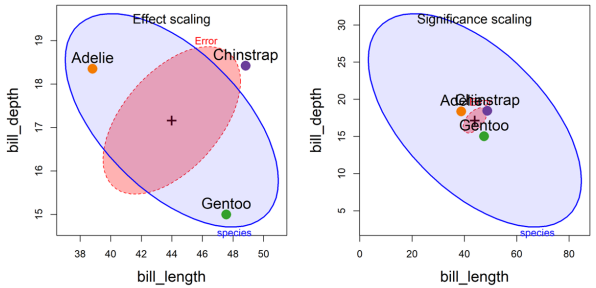
- The **E** ellipse is the **data ellipse of the residuals**
  - What you get when you subtract the group means from all observations, shifting them to the grand means.
  - **E** matrix is the "within-group **pooled** covariance matrix"

$$\mathbf{E} = \left(\hat{\mathbf{Y}} - \bar{\mathbf{Y}}\right)'\left(\hat{\mathbf{Y}} - \bar{\mathbf{Y}}\right) = \hat{\mathcal{E}}'\hat{\mathcal{E}}$$

50

50

HE plots: Effect scaling vs. Significance scaling



Effect scaling: Plot ellipses for H & E directly, centering

Direction of H vs. E shows how groups differ in a data-space view

Significance scaling: Scale H so an H ellipse that projects **outside** E signals a significant effect (Roy's test).

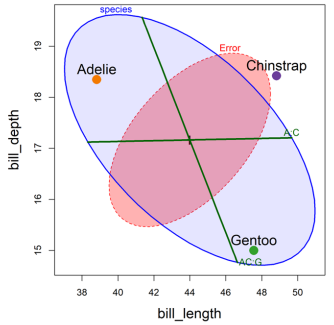
cf. Figure 12.5

51

51

Contrasts & linear hypotheses: What diff^{ces} make **H** big?

- Any linear hypothesis has its own **H**
  - A 1 df hypothesis is a line
- Orthogonal contrasts decompose the overall effect: $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2$
  - $\mathbf{H}_1$: (Adelie, Chinstrap) vs Gentoo
  - $\mathbf{H}_2$: Adelie vs Chinstrap
- These are conjugate axes of **H**
- Each "pushes" the **H** ellipse out in the direction of mean differences.



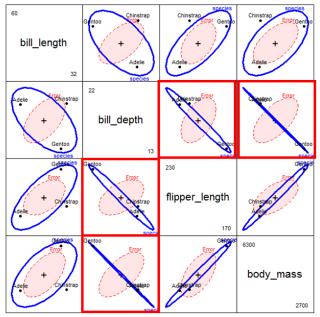
cf. Figure 12.7

52

52

HE pairs() plots

The `pairs()` method for mlm objects gives all pairwise HE plots in a scatterplot matrix format in data space.




```
pairs(peng.mod, size="effect",  …)
```

Something new here:

- avg. bill depth is negatively correlated with "size" variables – larger penguin species have smaller bill depths (curvature?)
- correlation of avg. bill depth with body mass nearly -1

53

53

Quantitative predictors: each effect is a line

- NLSY data: 243 children from the National Longitudinal Survey of Youth
  - Outcomes: reading, math
  - Predictors: father's income & education
- **Length**: the strength of the relationship
- **Direction**: how the predictor relates to the two responses
- **Overall test** is the **H** ellipse for all predictors jointly, H_0 : **B** = **0**

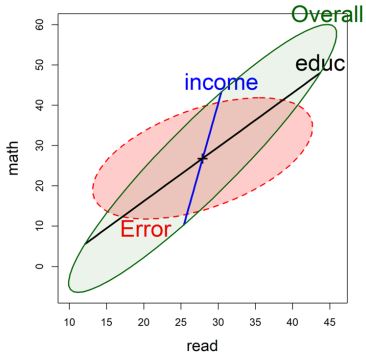


Figure 12.16

54

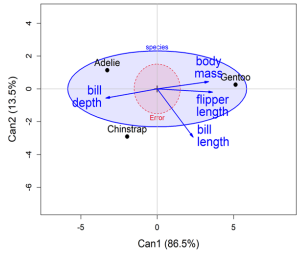
54

Canonical discriminant plots: The MANOVA juicer

- As with biplot, we can visualize MLM for all responses by projecting H and E into low-rank space
  - Canonical projection: $\boldsymbol{Y} \rightarrow \boldsymbol{Z} = \boldsymbol{Y}\boldsymbol{E}^{-1/2}\mathbf{V}$ where **V** = eigenvectors of HE^{-1}
- This is the view that maximally discriminates among groups
- CD HE plot: vectors show structure coefs (correlations) of variables w/ CD dimensions

Biplot of canonical scores

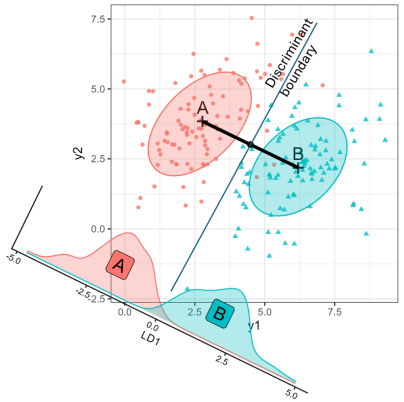
HE plot of canonical scores



55

55

Linear discriminant analysis: main ideas



Like a MANOVA flipped, so that group is the outcome, rather than a predictor.

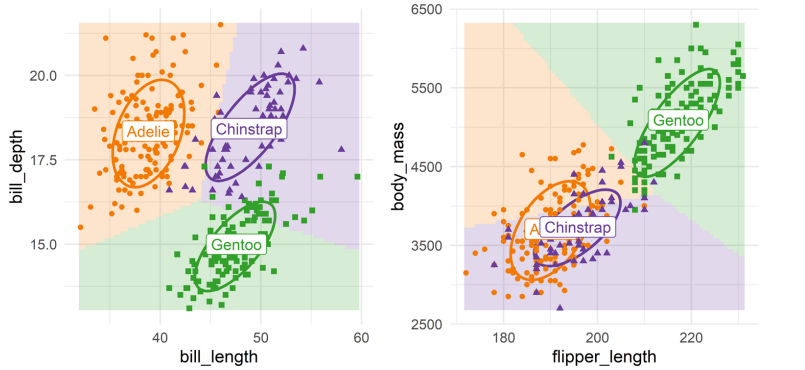
Emphasis is on prediction

Find the directions that best separate the groups, then classify by distance to the group means.

56

56

Prediction regions in data space



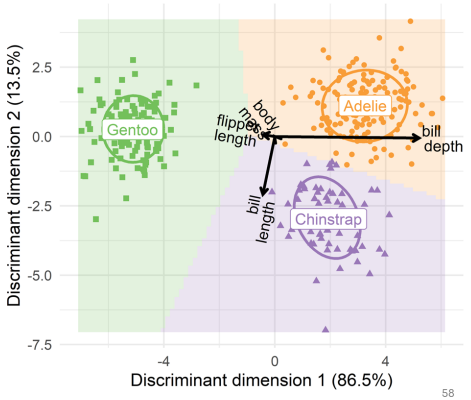
Classify every point in a grid of values: the LDA boundaries are linear.

57

57

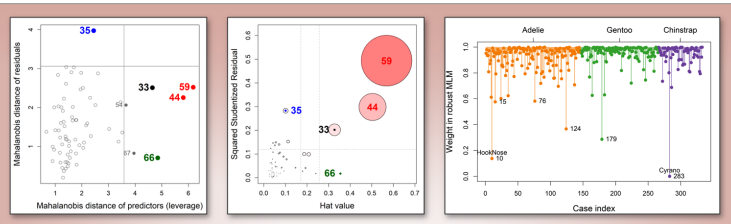
Biplot version of discriminant space

- With three species, two discriminant dimensions show everything
- Variable vectors show which/how measures separate the species



58

58



Checking multivariate models

Outliers, influence and robust estimation

Skip this ⏭

59

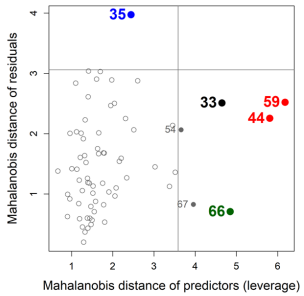
59

Unusual is not the same as influential

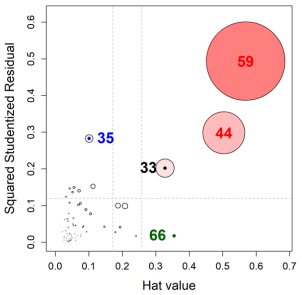
- School data: 70 schools, 3 outcomes, 5 predictors
- **Distance plot**: predictor distance (leverage) vs. residual distance (outliers)
- Lines: cutoffs for "large"

- **Influence plot**: bubble size ~ multivariate Cook's D
- **44, 59**: high leverage and influential
- **35**: an outlier, but little influence
- **66**: high leverage, but harmless

Distance plot: where is it unusual?



Influence plot: does it matter?



Refit without **44** and **59**: the same three predictors stay significant, but the reading and mathematics coefficients change by 77–104% for visit and over 300% for counseling.

Figure 11.18, 11.19

60

60

Who are the influential penguins?

- Multivariate influence plot
- The most influential case is our Chinstrap friend 'Cyrano'
- 'Snub Nose', an Adélie, is among the others
- One factor as predictor: hat values depend only on group size, so three columns

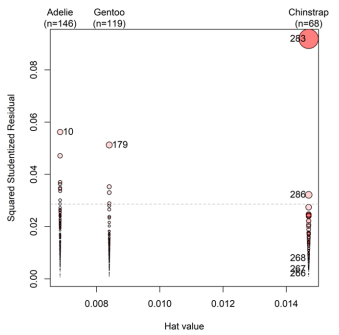


Figure 14.8

Robust MLMs: down-weight what does not fit

- robmlm(): iteratively reweighted least squares for a multivariate model
- Index plot of the final observation weights
- Cyrano gets weight 0; Snub Nose gets 0.13

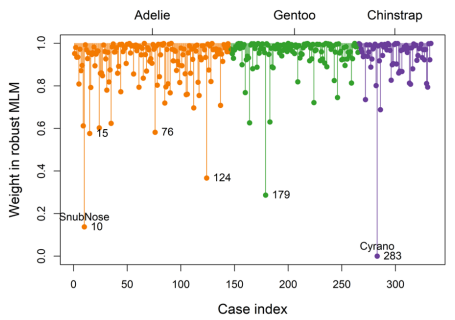


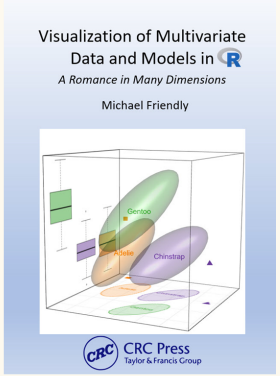
Figure 14.10

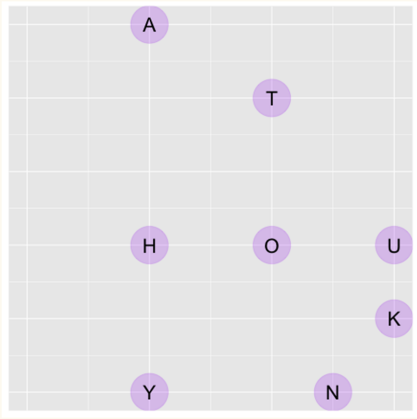
Summary: A few visual ideas go a long way

- **Ellipses**: for data, for coefficients, and for hypotheses and error
- **Visual thinning**: summarize, so the message can be seen
- **Projections**: every plot is a view; choose the views that answer the question
- **Geometry**: provides the connective tissue to understand relations in different, equivalent spaces.
- **Low-D views**: biplots and canonical plots show many variables in the space that matters
- Multivariate models deserve graphs as good as those for a single response

Book, code and packages

- Online version: friendly.github.io/Vis-MLM-book
- Source and R code: github.com/friendly/vis-MLM-book
- R packages: heplots, candisc, mvinfluence, genridge, VisCollin
- To appear with CRC Press / Chapman & Hall, 2027
- Blogs & random musings: https://friendly.github.io/blog/





Questions?
Discussion?

65

65